

Research Progress Report — Phases 6 and 7

Multi-LLM Comparison Platform: Comparative Evaluation of Code-Generation LLMs Across Web Development Task Categories

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1. Executive Summary

This report covers Phase 6 (Analysis and Recommendation Algorithm) and Phase 7 (Human Evaluation) of the Master's project. Phase 6 is complete. Phase 7 is fully instrumented and its protocol is finalized, but no human votes have been collected, so H3 remains untested.

Phase 6 required no new API calls. Every statistic and the entire recommendation algorithm were computed offline from the 325 evaluation runs collected in Phase 5. Before any new analysis was run, the published Phase 5 pass rates were re-derived from the stored data and reproduce exactly.

The phase produced four substantive results, three of which qualify conclusions drawn in the Phase 5 report.

First, H4 is rejected, and was arithmetically unreachable. The hypothesis predicted that task-aware routing would improve pass rate by more than 15% over a single-model strategy. Under leave-one-task-out cross-validation the oracle ceiling — a perfect router that always picks a model that passes — reaches only 58.5%, while always choosing GPT-OSS-120B already reaches 53.8%. The maximum improvement available to *any* routing algorithm on this dataset is therefore +8.6%, well below the 15% H4 required. Only three of 65 tasks are solvable by some model but not by GPT-OSS-120B.

Second, the recommendation algorithm as specified underperforms its own baseline, scoring 36.9% against 53.8% for always-best-single (−31.4%, bootstrap 95% CI [−27.7, −7.7] percentage points, $p = 0.0004$). This is a calibration failure, not an implementation failure: the Phase 2 framework fixed 30% of the scoring weight on latency and token efficiency before any data existed, which on this data steers selection toward fast, terse models and away from accurate ones. Removing those terms recovers performance to 53.8%, exactly tying the baseline.

Third, the Phase 5 model ranking does not survive statistical testing. Pairwise McNemar tests with Holm correction show GPT-OSS-120B is *not* distinguishable from Llama-3.3-70B ($p = 0.35$) or Llama-4-Scout ($p = 0.37$). The headline "GPT-OSS-120B achieved the highest overall pass rate" holds as a point estimate but not as a ranking claim.

Fourth, category-level specialization is weaker than reported. Of the five per-category tests supporting RQ1, only Bug Fixing reaches significance ($p = 0.002$). Frontend, API, CSS, and TypeScript do not, at 65 observations each.

Phase 7 instrumentation — a blind pairwise voting interface, a Bradley-Terry estimator, and a synthetic-vote generator — is built and validated. The estimator recovers planted model strengths to within 0.003. A synthetic

pilot of 1,200 votes was used to derive an evidence-based sample size for the human study (minimum 300 votes, target 800). These synthetic results validate the machinery only and are not evidence for H3.

2. Work Completed

2.1 Analysis Module (Phase 6 — Complete)

A new module, `pipeline/analysis/`, was built with five components:

File	Purpose
<code>data.py</code>	Single loading path for Phase 5 results; PostgreSQL with CSV fallback
<code>stats.py</code>	Chi-squared, Cramér's V, Spearman, McNemar, Holm correction
<code>recommender.py</code>	Two-stage task classifier and weighted scoring algorithm
<code>validate.py</code>	Leave-one-task-out cross-validation against three baselines
<code>run_analysis.py</code>	Orchestrator; writes JSON results and charts

All analyses run from one command (`python analysis/run_analysis.py`) and write machine-readable output to `pipeline/exports/analysis/phase6_results.json`, so every number in this report is generated rather than transcribed.

Test selection. Because every model attempts the same 65 tasks, model-versus-model comparisons are *paired*. McNemar's test is therefore correct and a two-sample proportion test is not. This distinction matters: it is the reason the GPT-OSS-120B advantage over Llama-3.3-70B fails to reach significance.

2.2 Recommendation Algorithm (Phase 6 — Complete)

The algorithm implements the formula fixed in the Phase 2 framework:

$$\text{Score}(\text{model}, \text{task}) = w1 \cdot \text{CategoryPassRate} + w2 \cdot \text{DifficultyPassRate} + w3 \cdot (1/\text{NormalizedLatency}) + w4 \cdot (1/\text{NormalizedTokens})$$

with the pre-registered defaults $w1 = 0.5$, $w2 = 0.2$, $w3 = 0.15$, $w4 = 0.15$. Latency and token terms are normalized by dividing by the best value across models, so the fastest model scores 1.0 and the terms are unit-free.

Stage 1 — task classification. Rule-based keyword matching. An early version counted keyword hits across all five categories at once and classified only 8% of bug-fix tasks correctly: a bug-fix prompt embeds the broken React or Express source, so it necessarily contains more language keywords than repair keywords. Bug fixing is an *intent*, orthogonal to language, so the classifier now tests repair intent first and lets it decide on its own; language categories compete only when repair intent is absent. This raised accuracy from 81.5% to 100% on the 65 benchmark prompts.

Stage 2 — profile lookup and scoring. Per-model empirical profiles built from training data only, then scored and ranked.

2.3 Validation Design (Phase 6 — Complete)

Leave-one-task-out cross-validation over all 65 tasks. For each fold, model profiles are rebuilt from the other 64 tasks; the held-out task never contributes to the profiles used to choose a model for it. An explicit assertion enforces this. The always-best-single baseline is likewise refitted per fold, so it enjoys no information advantage over the recommender.

Three baselines, as specified in the framework: random (exact expectation rather than sampled), always-best-overall, and oracle.

2.4 Human Evaluation Instrumentation (Phase 7 — Built, Not Yet Run)

A new top-level module, `human_eval/`:

File	Purpose
<code>app.py</code>	Blind side-by-side voting interface
<code>bradley_terry.py</code>	Maximum-likelihood Bradley-Terry fit with bootstrap intervals
<code>synthetic_votes.py</code>	Vote generator for pipeline validation
<code>run_synthetic.py</code>	End-to-end runner
<code>db.py</code>	Database access for the existing <code>human_votes</code> table
<code>PROTOCOL.md</code>	Full human evaluation methodology

No schema change was needed; the `human_votes` table has been in `db/schema.sql` since Phase 4.

Blinding. Model identity is never sent to the browser. The server maps an opaque comparison id to the two models and resolves it only when the vote is posted, so identity cannot be read from the page source. Left/right assignment is randomized per comparison.

Implementation note. The Phase 2 framework named FastAPI for the backend. The voting interface is built on the Python standard library instead. It is a research instrument used by a handful of evaluators, not the Phase 8 demonstration application, and removing the dependency removes an install step from every participant's setup. The only third-party import is `psycpg2`, already a pipeline dependency.

3. Results

3.1 Overall Model Differences (RQ1)

Model identity is associated with outcome across all 325 runs:

Test	Statistic	p	Cramér's V
Model $\times$ outcome (all tasks)	$\chi^2(4) = 27.04$	1.9×10^{-5}	0.288
Difficulty $\times$ outcome (pooled)	$\chi^2(2) = 86.06$	2.1×10^{-19}	0.515

Difficulty is a substantially stronger driver of outcome than model choice ($V = 0.515$ against 0.288). Which model you use matters less than how hard the task is.

3.2 Per-Category Differences (RQ1)

Category	$\chi^2(4)$	p	Cramér's V	Min expected count
Bug Fixing	16.78	0.0021	0.508	4.80
TypeScript	8.85	0.065	0.369	4.80
API	7.74	0.102	0.345	3.80
Frontend	3.65	0.456	0.237	3.20
CSS	0.93	0.921	0.119	4.80

Only Bug Fixing reaches significance. Every subtable has a minimum expected count below 5, so the chi-squared approximation is strained throughout and these p-values should be read as indicative.

This is a **weaker basis for RQ1 than the Phase 5 report claimed**. Category specialization is visible in the point estimates, but at 65 observations per category only the Bug Fixing effect is statistically distinguishable from noise.

3.3 Pairwise Model Comparisons (McNemar, Holm-corrected)

Model A	Model B	A only	B only	p (Holm)	Significant
GPT-OSS-120B	Qwen3-32B	28	1	< 0.0001	yes
Llama-3.3-70B	Qwen3-32B	22	1	< 0.0001	yes
Llama-4-Scout	Qwen3-32B	20	0	< 0.0001	yes
Llama-3.1-8B	Qwen3-32B	17	1	0.0010	yes
GPT-OSS-120B	Llama-3.1-8B	13	2	0.0443	yes
GPT-OSS-120B	Llama-3.3-70B	7	1	0.3516	no
GPT-OSS-120B	Llama-4-Scout	10	3	0.3691	no
Llama-3.1-8B	Llama-3.3-70B	1	6	0.3750	no
Llama-3.1-8B	Llama-4-Scout	2	6	0.5781	no
Llama-3.3-70B	Llama-4-Scout	5	4	1.0000	no

Five of ten pairs are significant, and four of those five involve Qwen3-32B. **The top three models are statistically indistinguishable from one another.**

3.4 Parameter Count and Performance (RQ2)

Spearman $\rho = 0.70$ between active parameter count and overall pass rate ($p = 0.19$, $n = 5$). The direction matches expectation — larger models perform better — but with five models this is descriptive only and cannot support an inferential claim.

3.5 Task Classifier

100% accuracy on all 65 benchmark prompts, with no misclassifications.

That figure is optimistic. Benchmark prompts share a rigid template ("BUG DESCRIPTION:", "Fix the bug in the following code"), which the classifier matches easily. Tested on ten freshly written paraphrases in natural developer phrasing, accuracy was **80%**. Both failures were bug reports lacking any explicit repair verb — "the submit handler fires twice on every click, sort it out" and "users report the cart total is off by one cent sometimes". Realistic classifier accuracy is therefore closer to 80% than 100%.

3.6 Recommendation Algorithm Validation (RQ3, H4)

Leave-one-task-out over 65 tasks:

Strategy	Pass rate
Oracle (upper bound)	58.5%
Always best single model (GPT-OSS-120B)	53.8%
Random selection	38.2%
Recommender (framework weights)	36.9%

Improvement over best single model: **−31.4%**. Bootstrap 95% CI on the difference: $[-27.7, -7.7]$ percentage points, $p = 0.0004$. **H4 is not supported.**

The headroom result is the more important finding. Oracle 58.5% against best-single 53.8% means the maximum improvement available to any router is **+8.6%**. H4 required more than +15%. The hypothesis was unreachable on this dataset regardless of algorithm design, because 35 of the 38 tasks solvable by any model are already solved by GPT-OSS-120B. Only **three tasks in the entire benchmark** are winnable exclusively by routing elsewhere.

Per category:

Category	Recommender	Best single	Random	Oracle
Bug Fixing	69.2%	84.6%	63.1%	84.6%
TypeScript	30.8%	61.5%	36.9%	69.2%
API	30.8%	46.2%	29.2%	46.2%
CSS	30.8%	46.2%	36.9%	53.8%
Frontend	23.1%	30.8%	24.6%	38.5%

3.7 Weight Sensitivity

w1 (cat)	w2 (diff)	w3 (lat)	w4 (tok)	Recommender	vs baseline
0.50	0.20	0.15	0.15	36.9%	−31.4%
1.00	0.00	0.00	0.00	53.8%	0.0%
0.70	0.30	0.00	0.00	52.3%	−2.9%
0.40	0.10	0.25	0.25	35.4%	−34.3%
0.25	0.25	0.25	0.25	36.9%	−31.4%

The result is driven entirely by the speed and efficiency terms. Any weighting that includes them underperforms; pure category weighting ties the baseline. No weighting tested beats it.

The weights were fixed in Phase 2 before data existed, and they were not tuned after the fact to rescue H4. This table is reported as a sensitivity analysis, not as a search for a passing configuration.

3.8 Zero-Test Sensitivity

Phase 5 scored 63 of 325 runs as failures despite the Docker harness reporting no test count at all. Re-running the analysis with those runs excluded:

Treatment	n	Overall χ^2 p	Cramér's V
All runs (as published)	325	1.9×10^{-5}	0.288
Zero-test runs dropped	262	0.369	0.128

The overall model effect loses significance under the alternative treatment.

This must be read carefully in both directions. Dropping those runs removes 40 of Qwen3-32B's 65 attempts, leaving badly unequal denominators:

Model	Passes / attempts (filtered)	Rate
GPT-OSS-120B	35 / 63	55.6%
Llama-3.3-70B	29 / 60	48.3%
Llama-4-Scout	28 / 60	46.7%
Llama-3.1-8B	24 / 54	44.4%
Qwen3-32B	8 / 25	32.0%

So the p-value moved partly because the association genuinely weakened and partly because most of one arm's data was deleted. **Neither treatment is correct.** Scoring a harness failure as a model failure is wrong when the harness was at fault; dropping it is wrong when the model emitted code so broken that no test could run. The honest conclusion is that the published Phase 5 result is **not robust to a defensible alternative treatment**, and resolving this requires the failure taxonomy that `failure_mode` was designed to capture but never recorded.

Because the filtered dataset is no longer a complete task × model grid, paired statistics on it are undefined. McNemar and the LOOCV validation refuse to run against it rather than silently treating missing cells as failures.

3.9 Phase 7 Synthetic Pilot

These results are synthetic. They validate the pipeline and are not evidence for H3.

Bradley-Terry fitted to 1,200 generated votes converged in 20 iterations:

Model	Rating	95% CI	Automated pass rate
Llama-3.3-70B	1028.7	[1007, 1055]	44.6%
GPT-OSS-120B	1027.2	[1006, 1052]	53.8%
Llama-4-Scout	1012.2	[986, 1035]	43.1%
Llama-3.1-8B	1003.7	[980, 1027]	36.9%
Qwen3-32B	928.1	[901, 954]	12.3%

Estimator check: Spearman $\rho = 1.000$ between fitted ratings and observed win rates, confirming the fit reproduces the ordering in the votes it was given. Independently, the estimator recovers planted strengths from a known generating distribution to within 0.003.

Correlation between simulated preference and automated pass rate is $\rho = 0.90$. This is the exact computation H3 calls for, but on synthetic votes it is circular — the votes were generated from those same pass rates. It is shown to demonstrate that the H3 machinery runs, nothing more.

The pilot also produced an evidence-based sample size for the human study (Section 4 of `human_eval/PROTOCOL.md`): 300 votes minimum to separate the extremes reliably, 800 for roughly ± 30 rating points per model.

4. Findings

1. **H4 is rejected, and was unreachable by construction.** Maximum available improvement from perfect routing is +8.6% against the +15% required. Only 3 of 65 tasks are winnable by routing away from the single best model. Any future routing hypothesis must first establish that headroom exists.
2. **Routing headroom requires model complementarity, which this model set lacks.** GPT-OSS-120B solves 35 of the 38 tasks any model solves. Routing pays off only when different models succeed on *different* tasks; here their successes are nearly nested. A future model set should be selected for complementarity, not only for architectural diversity.
3. **Pre-registered weights were miscalibrated for this regime.** Spending 30% of the score on latency and token efficiency costs 17 percentage points of accuracy. Efficiency terms make sense when accuracy is comparable across models; here the accuracy spread is large enough to dominate.
4. **The Phase 5 ranking is a ranking of point estimates, not of models.** The top three models are statistically indistinguishable. Only the Qwen3-32B deficit and the GPT-OSS-120B advantage over Llama-3.1-8B survive correction for multiple comparisons.
5. **Difficulty dominates model choice.** Cramér's V of 0.515 for difficulty against 0.288 for model. For a practitioner, task complexity predicts success better than model selection does.
6. **Nineteen percent of the Phase 5 dataset rests on an unresolved scoring decision.** The overall model effect is significant under the published treatment and not significant under a defensible alternative. This is the most consequential outstanding issue in the project.
7. **Rule-based task classification is brittle.** 100% on templated benchmark prompts, 80% on natural paraphrases, with both failures on bug reports lacking explicit repair verbs. A learned classifier is the appropriate Phase 8 upgrade.

5. Challenges and Design Decisions

1. **Bug-fix classification collapse.** The initial single-pass keyword classifier scored 8% on bug-fix tasks because bug-fix prompts embed source code in the language of another category. Resolved by treating repair intent as orthogonal to language and testing it first. A false-positive audit against all 65 prompts then removed two over-broad markers ("debug" alone, which fires on a TypeScript prompt merely mentioning debugging, and "instead of", which fires on a design prompt).
2. **A silent pandas defect in contingency-table construction.** Selecting columns with `table[[False, True]]` on a table with boolean column labels is interpreted by pandas as a boolean *row* mask, not column selection, and silently returned a single-row table. Caught by a sanity check asserting that a table with no association yields $p \approx 1$. Fixed by using explicit `reindex`. The episode is the reason every statistical function is now validated against distributions with known answers before being run on real data.

3. **NaN coercion inflating the oracle.** On the filtered dataset the task $\times$ model grid becomes ragged, and calling `.astype(bool)` on the pivot turned every missing cell into `True`, inflating the oracle bound from 38/63 to 56/63 and corrupting the first zero-test sensitivity figures. Fixed by filling explicitly and by adding guards that refuse to compute paired statistics on an incomplete grid.
4. **Synthetic preference model over-weighted concision.** The first generator applied a concision tiebreak whenever both models passed or both failed. Since most task/model pairs fail, that tiebreak decided the majority of comparisons and ranked the most accurate model fourth. The weight was reduced and split (0.15 when both pass, 0.05 when both fail) on the reasoning that two broken solutions give a reviewer very little to judge. The accompanying validation check was also corrected: it compared fitted ratings against *pass rates*, which conflates whether the estimator works with whether the generator agrees with pass rate. It now compares against the empirical win rates in the generated votes.
5. **Framework deviation on the web stack.** FastAPI was specified but is not installed in the project environment, and adding it would modify the environment for a component that does not need it. The voting interface uses the standard library instead.

6. Current Status and Next Steps

#	Phase	Period	Status	Output
1	Literature Review	Feb–Mar 2026	Complete	16 sources
2	Framework Design	Mar–Apr 2026	Complete	v2.0 doc
3	Build Benchmark Suite	Apr 2026	Complete	65 tasks
4	Build Evaluation Pipeline	Apr 2026	Complete	7 components
5	Run Experiments (Run 1)	Apr 2026	Complete	325 data points
6	Analysis & Algorithm	May–Jul 2026	Complete	5 modules, H4 tested
7	Human Evaluation	Jul 2026 –	Instrumented; collection pending	Protocol + tooling
8	Build Application	Aug 2026	Planned	
9	Thesis & Defense	Aug–Sep 2026	Planned	

Immediate next steps

1. **Recruit evaluators and collect human votes.** Everything needed is built. Target 800 votes from 8–12 evaluators. Synthetic votes must be cleared first (`python run_synthetic.py --clear`).
2. **Resolve the zero-test rows.** This now blocks a clean thesis claim. Two parts: confirm whether API-013 and CSS-012 are defective tasks by running them against their reference solutions in Docker, and populate `failure_mode` so harness failures can be separated from model failures.

3. **Re-run Qwen3-32B at a higher token limit.** Thirteen of its 65 runs hit the 4,096-token ceiling and ten of those produced no test count. Until this is separated, the Phase 5 conclusion that chain-of-thought reasoning degrades code generation is confounded with simple output truncation.
 4. **Pass@3 runs**, subject to Groq's 100K tokens/day/model limit. Single-run evaluation remains the largest threat to every result in this report.
 5. **Reconsider the routing question for Phase 8.** Given +8.6% headroom, a demonstration application framed around "which model should I use" has little to demonstrate. Two better framings: select a model set chosen for complementarity, or reframe the application around *difficulty-aware* selection, since difficulty is the stronger signal.
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7. Threats to Validity

Carried forward from Phase 5: single-run evaluation (Pass@1 only), model substitution relative to the original framework, single inference provider (Groq LPU), fixed temperature 0.2, and benchmark scope of 65 tasks across five categories.

New to this phase:

- **Zero-test treatment.** The central RQ1 result is significant under the published treatment and not significant when the 63 zero-test runs are dropped. Unresolved, and the most serious threat in the project.
 - **Statistical power.** At 65 observations per category, only a large effect is detectable. The four non-significant category results are consistent with genuine specialization that this sample cannot resolve; absence of evidence is not evidence of absence.
 - **Chi-squared assumptions.** Every per-category subtable has a minimum expected count below 5. Those p-values are indicative rather than exact.
 - **n = 5 models.** The Spearman correlation between parameter count and pass rate, and the eventual H3 correlation, both rest on five points.
 - **Validation is within-benchmark.** LOOCV holds out tasks but not the benchmark. Estimated recommender performance would not transfer to tasks drawn from a different distribution.
 - **Classifier accuracy is overstated by the benchmark.** 100% on templated prompts against 80% on natural paraphrases; the deployed figure is the latter.
 - **Synthetic votes are not human data.** No claim in this report about human preference is supported by evidence. H3 is untested.
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8. Reproducing These Results

```
cd pipeline
source venv/bin/activate
python analysis/run_analysis.py          # all Phase 6 statistics and charts

cd ../human_eval
python run_synthetic.py                 # Phase 7 pipeline validation
python app.py --port 8000                # voting interface
```

Outputs are written to `pipeline/exports/analysis/phase6_results.json`, `pipeline/exports/analysis/phase7_synthetic.json`, and `pipeline/exports/charts/phase6_*.png`. These paths are git-ignored, which is why every figure in this report is reproduced inline.

9. References

[1] K. Xu et al., "Web-Bench: A LLM Code Benchmark Based on Web Standards and Frameworks," arXiv:2505.07473, May 2025. [2] W. Chiang et al., "Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference," arXiv:2403.04132, March 2024. [3] LMArena, "WebDev Arena: A Live LLM Leaderboard for Web App Development," 2024–2025. [4] "Learning to Route LLMs with Preference Data," ICLR 2025. (RouteLLM) [5] "Query-Based Router by Dual Contrastive Learning," NeurIPS 2024. (RouterDC) [6] "A Graph-based Router for LLM Selections," ICLR 2025. (GraphRouter) [7] EvalPlus: Rigorous Evaluation of LLM-Synthesized Code, github.com/evalplus/evalplus, 2024. [8] C. Jimenez et al., "SWE-bench: Can Language Models Resolve Real-World GitHub Issues?" 2024. [9] R. A. Bradley and M. E. Terry, "Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons," *Biometrika* 39(3/4), 1952. [10] D. R. Hunter, "MM Algorithms for Generalized Bradley-Terry Models," *Annals of Statistics* 32(1), 2004. [11] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika* 12(2), 1947. [12] S. Holm, "A Simple Sequentially Rejective Multiple Test Procedure," *Scandinavian Journal of Statistics* 6(2), 1979. [13] Groq, "Supported Models," <https://console.groq.com/docs/models>, accessed April 2026. [14] LiteLLM Documentation, <https://docs.litellm.ai/docs/>, accessed April 2026.